

MD ARAFAT HOSSAIN

Staff Engineer | iOS, macOS

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CAREER SUMMARY

Experienced Software Engineer with 13+ years building production-grade applications across iOS, macOS, Windows, and IoT platforms. Contributed to major products such as **SmartThings iOS** (62M+ users) and served as a lead developer for **Samsung DeX** on macOS and Windows, driving architecture, feature development, and performance enhancements. Skilled in Swift, Swift Concurrency, UIKit/SwiftUI, MVVM/MVVM-C, and scalable architecture design, with additional working experience in Objective-C, WPF, MFC, and system-level integration. Known for delivering high-quality solutions, improving legacy codebases, and supporting engineering teams through complex development challenges.

TECHNICAL SKILLS

iOS Development

- Swift & Swift 6 (Concurrency – async/await, actors), UIKit, SwiftUI, AutoLayout, Storyboards, XIB/NIB, Objective-C
- **Combine**, ReactiveSwift, architectural patterns: MVVM, MVVM-C, Redux
- RESTful API integration
- Concurrency and thread-safe state management (async/await, actors)
- Unit Testing: XCTest, basic Quick/Nimble
- Performance profiling using Time Profiler
- macOS application development (Samsung DeX for PC – architecture & release)

Other Technologies

- Git, **GitHub**, Perforce, GitHub Actions, Jenkins
- C/C++, C#, **Java**, JUnit (Java unit testing), WPF, MFC
- Basic Android development (Java)
- **Agile / Scrum** methodologies

WORK EXPERIENCE

Samsung R&D Institute Bangladesh Ltd. | Staff Engineer-I | Mar 2021 – Present

SmartThings iOS

- Led development of **Settings** and **MapView** features for **SmartThings iOS** (62M+ active users, 4.5+ rating).
- Integrated new device cards (TV, light, sensors) and enhanced MapView experiences.
- Migrated modules from **ReactiveSwift** → **Combine** and modernized architecture.
- Migrating existing modules to **Swift 6 Concurrency** (async/await, actors).
- Implemented **Do Not Disturb** and **Emergency Alert** settings.
- Conducted performance tuning using Time Profiler and resolved major **VoCs**.

Godot Migration for MapView 3D

- Replacing Unity with [Godot](#) for 3D rendering.
- Contributing to [SwiftGodot](#) & [SwiftGodotKit](#) open-source integration.

Other Key Contributions

- Modernized UI and improved **IPC** for 5D Viewer (WPF) used in medical ultrasound systems.
- **Tech Lead** for **Samsung DeX for PC** (7M+ downloads, 757K+ monthly users) — planning, architecture, optimization.
- Performed memory improvements and code reviews on MFC-based Windows apps.
- Achieved **Professional** Level (Top 20%) in Samsung's **SW Certificate** Test.

Samsung R&D Institute Bangladesh Ltd. | Engineer-III | Mar 2019 – Feb 2021

- **Lead Developer & Release** (Manage [Notarization](#)) **Maintainer** of Samsung DeX for PC (macOS).
- Delivered universal macOS app support for **Apple M1**.
- Implemented USB & wireless connectivity between Android devices and macOS.
- Built file transfer, RTSP-based screen rendering, and input-sharing features.
- Ensured performance, stability, and cross-platform compatibility.
- Awards:
 - Samsung Mobile R&D CTO Award (2019)
 - SRBD Best Team Award (2019)

Samsung R&D Institute Bangladesh Ltd. | Engineer-II | Mar 2017 – Feb 2019

- Built **UI Automator** for Android & Tizen, enabling internal B2B device testing.
- Developed an iOS UI testing framework similar to Appium for automation.
- Created an Eclipse plugin to generate automated test cases and integrate internal tools.

Samsung R&D Institute Bangladesh Ltd. | Engineer-I | Jun 2013 – Feb 2017

- Contributed to [IoTivity SDK](#), adopted by Samsung, LG, and Microsoft IoT solutions.
- Developed [OCF](#)-compliant APIs enabling cross-platform IoT interoperability.
- Built a C# .NET desktop application to remotely control Android & Tizen devices.
- Achieved **Advanced** Level (Top 50%) in Samsung's **SW Certificate** Test.

RTC Hubs Limited | Junior Programmer | Sep 2012 – Jun 2013

- Created Multi Shot Cam, a top-charted iOS app generating highest revenue at RTC Hubs.
- Developed 8+ iOS apps including camera & productivity tools using [GPUImage](#).

MENTORING AND LEADERSHIP

SmartThings (iOS) – Settings & MapView | Mar 2024 – Present

- **Part Lead** for Settings & MapView; owned roadmap, design, and delivery for a **6-member** team.
- Mentored engineers on MVVM/Combine, Swift/SwiftUI/UIKit standards, and development workflows.
- Conducted training on Swift Concurrency, scalable architecture (MVVM, MVVM-C, Redux, Combine), and IoTivity fundamentals.
- Coordinated responses to Market VoCs, resolved high-priority issues, and aligned with QA/Product.

Samsung Medison – 5D Viewer | Jan 2023 – Feb 2024

- **Tech Lead** overseeing documentation, cross-team collaboration, and knowledge sharing.
- Facilitated onboarding and architecture clarification for distributed teams.

Samsung DeX for PC | Mar 2019 – Dec 2022

- **Tech Lead** for Windows/macOS/Android teams; ensured high-quality feature delivery.
- Mentored **12+ engineers** and guided architecture, performance, and release pipeline.
- Delivered PoC to Samsung CTO securing executive approval.
- Collaborated with PMs and VPs to define functional/non-functional requirements and drive product direction.

Organization-wide Contributions

- Onboarded 7+ engineers across multiple projects.
- Participated in hiring—technical interviews & evaluations.

EDUCATION AND CERTIFICATION

B.Sc. in Computer Science and Engineering

Rajshahi University of Engineering & Technology

Feb 2008 – Sep 2012